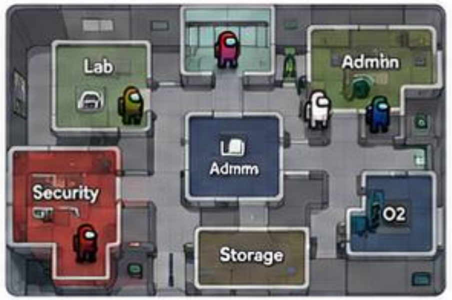
Game World Design Direction

Phantom Q Headquarters as a shared cybersecurity environment

The Sims

- Human spatial interaction
- Objects and workstations
- Room-based interactions
- Character movement

Among Us



- Readable locations
- Restricted areas
- Short routes
- Location-based suspicion

Phantom Q Headquarters

Human environment

- Relatable characters
- Believable workplace

Readable map

- Distinct rooms
- Clear routes

Security context

- Where actions occur matters
- Who is authorised matters

Phantom Q combines human spatial interaction with security-focused location awareness rather than copying either game directly.

Raised-Oblique 2.5D Camera

Balancing room readability, character visibility and development complexity

Top View 90°

Too flat

Side / Low 3D 20–30°

Too much environmental detail required

Raised Oblique 55–65°

Best balance

Camera rules

- Fixed orientation
- Limited zoom
- Soft player follow
- No free rotation in Release 1
- Temporary close-up for interactions

A fixed raised-oblique view provides spatial depth without creating the complexity of a fully free 3D world.

Camera Behaviour by Interaction

One world view with controlled changes in visual focus

Headquarters View

- Shows several rooms
- Supports navigation
- Displays facility status

zoom in →

← zoom out

Room View

- Focuses current location
- Nearby interactions visible
- Characters and objects clearer

zoom in →

← zoom out

Interaction View

- Workstation, USB, doors or equipment
- Used during active tasks
- Returns to world view when complete

System-selected focus

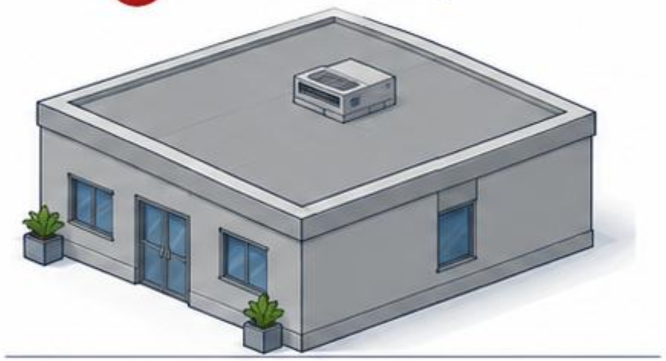
The learner does not manually manage the camera. The system chooses the useful view for the current task.

Camera changes should support the learner's task rather than becoming another control the learner must master.

Room Visibility & Cutaway Architecture

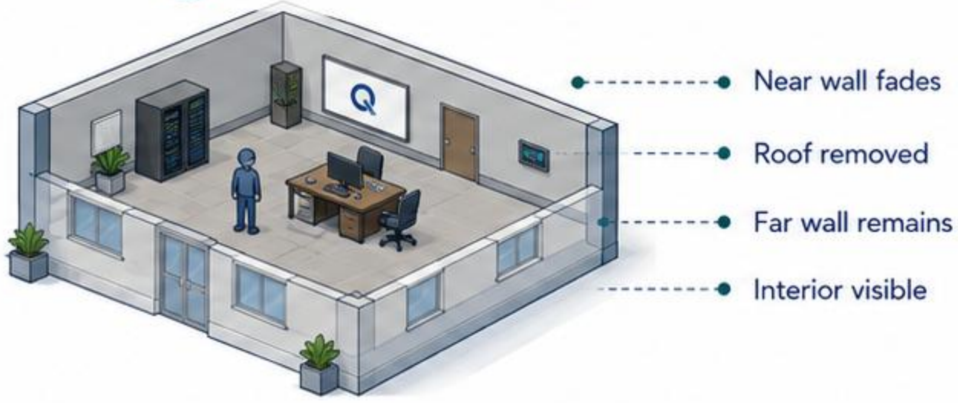
Keeping interiors readable from the raised-oblique camera

Normal building



- Walls block player
- Interior interactions hidden

Phantom Q cutaway



1. Player approaches room
2. Roof fades
3. Front wall lowers
4. Interior becomes visible

Why this matters

- Preserves architectural feel
- Keeps desks and equipment visible
- Supports USB, workstation and QKD interactions

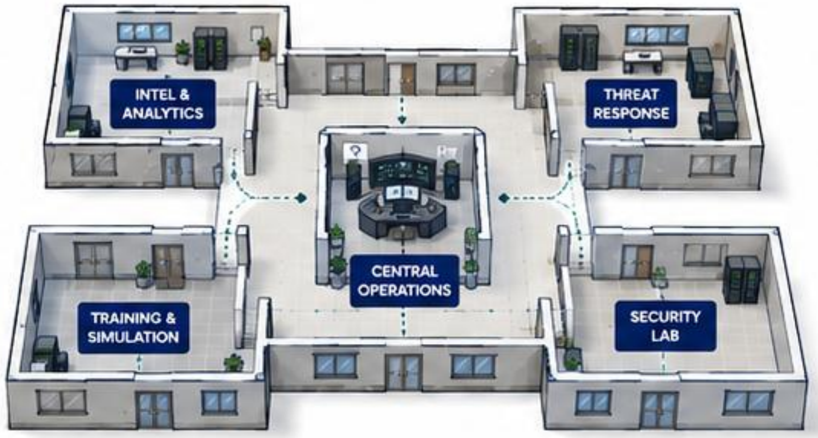
The Headquarters should feel architectural while remaining as readable as a game board.

Headquarters Spatial Design Rules

How the facility remains understandable while still feeling like a real organisation

1 Central anchor

Central Operations is always recognisable.



2 Short wings

Important facilities connect through short paths.

3 No maze

Corridors guide rather than confuse.

4 Restricted areas remain visible

Learners can see future capability before unlocking it.

5 Location must mean something

Rooms correspond to actual security functions.

Bad

- Long corridors
- Identical rooms
- Hidden destinations
- Decorative spaces

VS

Good

- Clear landmarks
- Distinct facilities
- Short routes
- Security meaning

Navigation should create context, not difficulty.

Headquarters Functional Zones

Grouping facilities by what learners actually do inside them

LEARN

- Training & Foundations
- Cryptography

OPERATE

- Central Operations
- Communications

Central HQ

ATTACK / TEST

- Red-Team Lab
- Advanced Compute

MONITOR / RESPOND

- SOC
- Secure Core

ENGINEER

- Engineering Workshop
- Quantum Wing

This is the functional relationship map, not yet the floor plan.

Every Headquarters area must represent an actual learning or operational capability.

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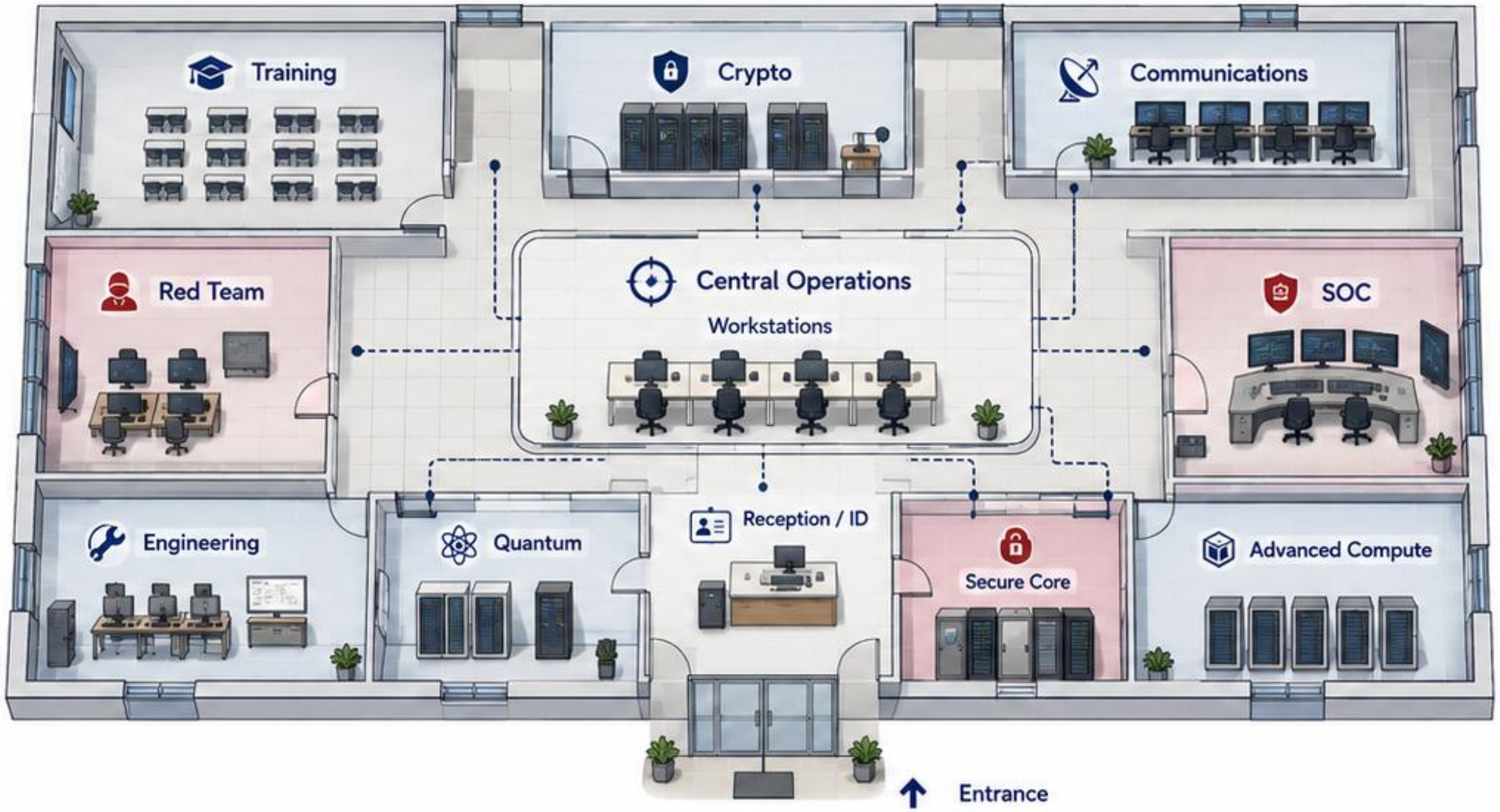
Release 1 Headquarters — Functional Floor Plan

A single-level environment organised around Central Operations



Open areas
General access for all personnel.

Visible restricted areas
Controlled access with clear visibility.

Classified / deeper access areas
Strict access controls and limited visibility.

Release 1 uses one readable headquarters level rather than multiple floors or an open-world campus.

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Central Operations — The Player Anchor

The space players repeatedly return to throughout the experience

Home location

The default hub players return to regularly.

Orientation point

Clear landmarks and sight lines for instant reorientation.

Basic workstation location

Access desks, tools and systems from a consistent spot.

Common meeting point

A natural place for players to gather and coordinate.

Incident and status reference

Real-time overview of headquarters activity and global incidents.

Clear routes to other wings

Visible exits and signage keep navigation simple and intuitive.

Players should be able to reorient themselves immediately whenever they return to Central Operations.

Character Movement Model

Combining direct movement with intelligent object interaction

1 FREE MOVEMENT

- Desktop
- WASD / arrow keys
- Touch / virtual joystick
- Optional tap destination

2 CONTEXTUAL INTERACTION

1 Select workstation → 2 Character finds route → 3 Walks to correct position → 4 Interaction becomes available

BAD: Precision clicking

- Pixel-perfect movement
- Constant micro-adjustments
- Long setup to use objects
- Creates friction and breaks flow

GOOD: Intelligent interaction

- Approach object
- Auto-position
- Interact naturally
- Keeps focus on strategy, not controls

Free movement creates presence in the world.

Players can explore and navigate freely using familiar controls.

Contextual pathing prevents precision-walking frustration.

The game handles positioning so players focus on decisions, not micromanagement.

No precision walking should be required.

Interactions should feel natural, reliable, and effortless.

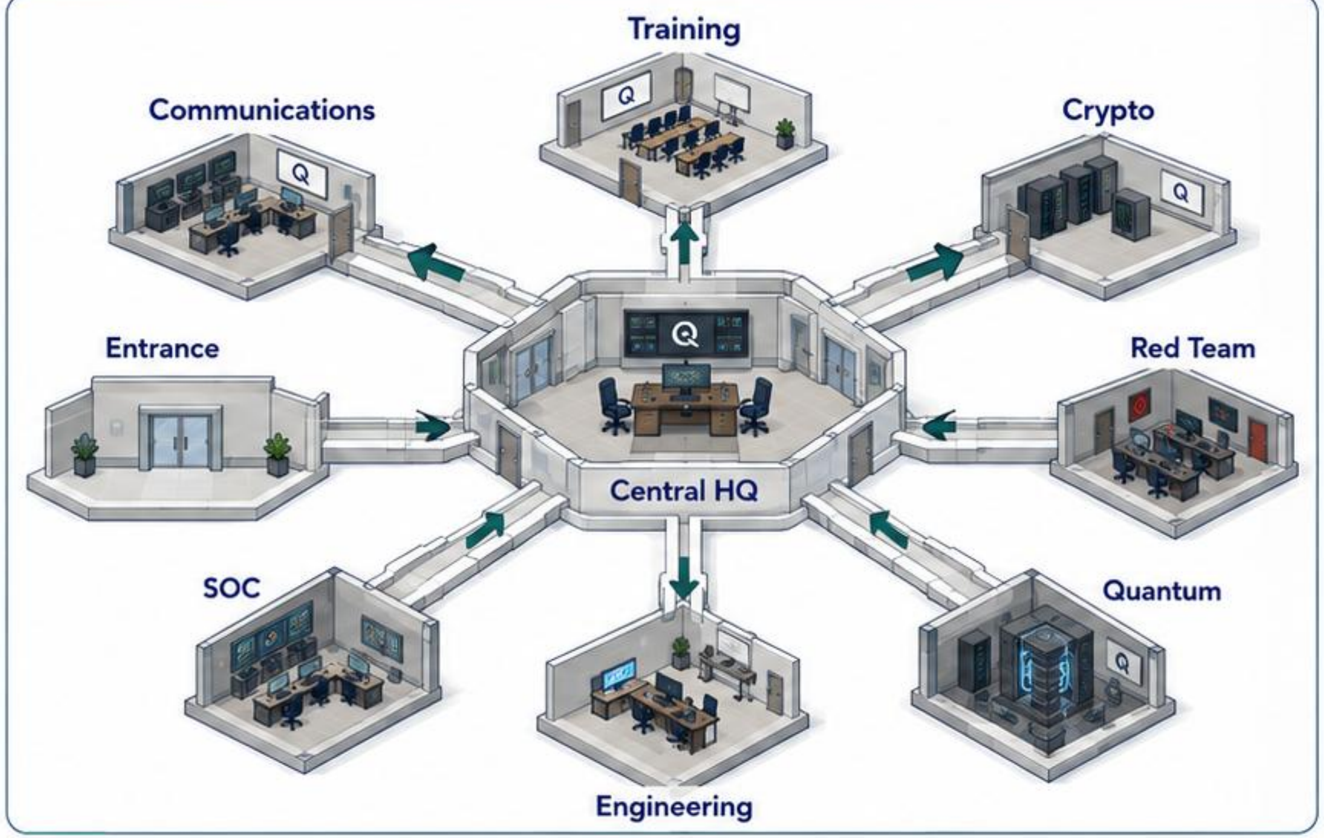
Movement creates presence; intelligent interaction prevents movement from becoming frustrating.

Headquarters Circulation Network

Designing paths around security context rather than travel distance

Security Depth

- Higher-security areas should feel deeper
- Use checkpoints, doors and restricted corridors
- Do not rely on huge walking distances



Route Logic

- Central → Communications = short
- Central → SOC = short
- Central → Training = short
- Advanced facilities slightly deeper

Path Design Comparison

BAD

Long walking times, maze-like corridors, confusing detours.

GOOD

Clear paths, fast movement, visible access points, layered security.

Security depth should come from access control, not from forcing players to walk for minutes.

Doors, Access & Clearance

Making clearance visible inside the environment

OPEN

Communications

- Access granted
- General access.

RESTRICTED

SOC

- Clearance 2 required
- Controlled access.

HIGH SECURITY

Quantum Wing

- Clearance 4 required
- Specialised access.

CLASSIFIED

Secure Core

- Access classified
- Highest security level

What a door communicates

- Facility name
- Required clearance
- Current lock state
- Incident state
- Authorised users

Access control becomes something the learner physically encounters, not an abstract menu permission.

Location Becomes Security Context

Turning movement through Headquarters into usable cyber evidence

CHARACTER MOVEMENT
A person moves through Headquarters.

→

LOCATION EVENTS
The system captures precise location events.

→

ACTIVITY RECORD
Events are structured into an activity record.

→

SOC / INCIDENT EVIDENCE
SOC uses movement-linked evidence in investigations.

EVENT FIELDS (Captured for every event)

WHO? | WHAT? | WHERE? | WHEN? | AUTHORISED?

	WHEN	WHO	WHERE	WHAT	AUTHORISED?
EXAMPLE A	09:04	Trainee_01	Central Operations	Login	AUTHORISED
EXAMPLE B	09:17	Trainee_01 credential	Unknown workstation	Login	SUSPICIOUS

The map is part of the cybersecurity evidence model.

Location is not only for navigation.

SOC can use movement-linked evidence in investigations.

The map is not only navigation; it becomes part of the cybersecurity evidence model.

Character Art Direction

Human enough to feel relatable, stylised enough to remain practical

Photorealistic

Extreme cartoon

Among Us-like body proportions

Stylised human

Target appearance

- Realistic proportions
- Simplified facial detail
- Readable hairstyles
- Readable clothing
- Recognisable silhouettes
- Efficient for multiple devices

Use

- Relatable characters
- Practical 3D production
- Works at oblique camera distance

Characters should feel like people working inside Phantom Q, not game tokens or photorealistic actors.

South African Representation

Building a cast that naturally reflects the environment in which Phantom Q is being developed

Natural variation

- Skin tones
- Hairstyles
- Body types
- Ages
- Professional style

Important rule

Role ≠ ethnicity

Representation principles

- Diversity should appear naturally across the cast
- Ethnicity is not a gameplay mechanic
- Roles are not assigned by appearance
- Character quality should remain consistent across the full roster

Design intent

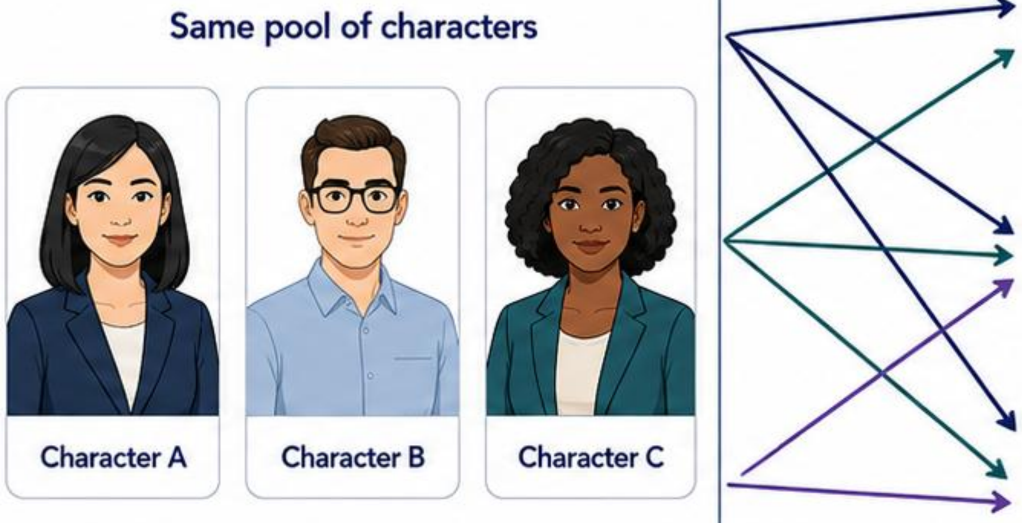
- Reflect South African reality
- Support player identification
- Avoid stereotypes

Representation belongs in the character system naturally; ethnicity is never a security role or gameplay attribute.

Character Identity vs Security Role

Keeping Alice, Bob and Eve as operational perspectives rather than fixed people

Same pool of characters



ALICE ROLE
Operational perspective and responsibilities

BOB ROLE
Operational perspective and responsibilities

EVE ROLE
Operational perspective and responsibilities

Role communicated through

- Badge
- Workstation assignment
- Small UI marker
- Authorised tools
- Objectives

Not communicated through

- Ethnicity
- Permanent costume
- Villain appearance

Eve should not visually look evil; she represents an operational perspective and can appear fully professional.

Why this matters

- Supports role rotation
- Avoids stereotype bias
- Keeps identity and gameplay flexible

The person remains the person; Alice, Bob and Eve describe what security responsibility they currently hold.

Game World Architecture Summary

How Headquarters, characters, movement and security context work together

World

- One headquarters
- Fixed raised-oblique camera
- Cutaway architecture

Movement

- Direct + contextual
- Short routes
- No maze

PHANTOM Q HEADQUARTERS

↓

RAISED-OBLIQUE 2.5D WORLD

↓

CHARACTERS | ROOMS | OBJECTS

↓

MOVEMENT | LOCATION | INTERACTION

↓

SECURITY EVENTS

↓

WHO / WHAT / WHERE / WHEN / AUTHORISED?

↓

SOC / INCIDENTS

Characters

- Stylised humans
- Diverse South African representation
- Roles separated from identity

Security

- Location matters
- Access matters
- Movement can become evidence

The Headquarters is simultaneously the game world, navigation structure and security context for Phantom Q.